Response to the review of "Getting the Surface Right"

Revised manuscript: `paper/full_paper_revised.pdf` (built from the source fragments by `paper/build_full_paper.py`). The original `full_paper.pdf` is unchanged. 3 October 2026.

Summary

We checked every finding against the code, saved outputs and sources; all 40 were confirmed or reasonable, and we disagree with none. The revision makes the following substantive changes.

- **New computation without training:** corrected uncertainty benchmark with signals independent of the stereo mask (F01); disjoint regions with error shares from per-frame sums (F10, F24); metric and fixed-scale depth errors and per-frame scale statistics for all 24 trained models (F02); prior-only evaluation (F31); training-frames-only scale estimation with known-pose geometric filters and a sensitivity analysis (F06, F07); unrounded statistics and per-seed paired differences (F13, F27, F30).
- **New training:** our model retrained (3 seeds) with the held-out-clean, filtered scale $\times 0.104$ instead of $\times 0.093$: 32.29 ± 0.06 dB PSNR and 0.147 scale-aligned AbsRel (vs. 32.40 and 0.148 with $\times 0.093$; per seed -0.08 to -0.16 dB). The depth result does not depend on the test frames; the PSNR gain over the reproduced baseline becomes $+0.31$ dB.
- **Reframing without new training**, as the review's minimum revision allows: confounded ablations (F04, F05), second-sequence scope (F11), metric claims (F02), protocol separation (F03).
- **Text, equations, figures and build:** all remaining findings.

Effect on the conclusions. The uncertainty conclusions hold after removing the mask dependence (prior disagreement 34–65 % of the attainable gain instead of 30–63 %; accumulated opacity unchanged at -1 to 17 %). Per-frame alignment hides little temporal drift (one training-view scale gives at most 0.003 higher AbsRel), but our P1 reconstructions are not metric (raw depth $1.87\times$ too shallow relative to stereo), so all depth claims are now stated as shape agreement with an estimated reference. The trained models' depth largely inherits the prior's accuracy (new prior-only Table V), and depth edges are not the dominant source of total error (tissue carries 63 % on P1).

Point-by-point

ID	Status	Change
F01	Accepted; recomputed	Prior alignment now uses only pixels where rendered and prior depth are finite and positive; all signal maps are frozen before any stereo mask (`stereomis/uncertainty_bench.py`; the original kept as `uncertainty_bench_v1_reference_masked.py`). Table VIII (formerly VI), Figs. 7–10 and all quoted ranges regenerated from `output/stereomis/uncertainty_v2`. Conclusions unchanged; prior-disagreement gain 34–65 %.
F02	Accepted; new evaluation	Depth metrics renamed "per-frame scale-aligned, stereo-reference-relative". New Table VI: aligned, fixed-scale (one factor from training views) and metric AbsRel plus per-frame scale mean and CV for all setups (`stereomis/eval_depth_extended.py`). Fixed vs. aligned differ by ≤ 0.003; scale CV 0.4–6.2 %; ours is $1.87\times$ too shallow (metric AbsRel 0.462). "Defined only up to scale" removed; metric claims narrowed.
F03	Accepted	P1 (monocular + known poses) and P2_5 (stereo-assisted scale) are separated in the problem setting, Fig. 1, Table I, Table VII (two blocks), the setup and the conclusion. The ten P2_5 calibration frames are listed; none is a test frame. The unscaled P2_5 row is kept as the monocular comparison.
F04	Accepted; reframed	The normal-loss block now states its stereo-fitted scale and includes its matched reference (31.56); differences reported as $-0.12/-0.28$ dB.
F05	Accepted; reframed	The endpoint is named "$\lambda_d{=}\lambda_n{=}\lambda_u^D{=}0$: all depth terms and image TV off"; the text states that removing the group improves PSNR without identifying individual terms. Leave-one-out runs not added (listed as future work).

ID	Status	Change
F06	Accepted; recomputed	Quasi-static assumption stated. Added known-pose Sampson filter ($\leq$2 px), cheirality in both cameras, reprojection ($\leq$2 px) and triangulation-angle filters, and a baseline sensitivity (2/4/6 mm: 0.085/0.104/0.105) (`stereomis/estimate_prior_scale.py`).
F07	Accepted; recomputed and retrained	Confirmed that 0.0928 used all frames. Training-frames-only: 0.0976; with filters: 0.1044. The paper states which factor each experiment used and reports the retrained model with 0.1044.
F08	Accepted	"True error"/"ground truth" replaced by stereo-reference-relative terminology; the strict-reference claim weakened to "persists after filtering some stereo mismatches"; region coverage under the strict mask reported.
F09	Accepted	Signal renamed "accumulated opacity (used as confidence by Endo-4DGS)"; the loss written with its global reductions; the finding scoped to this signal and loss.
F10	Accepted; recomputed	Disjoint partition; error shares from per-frame sums: tissue 63 %, edges 22 %, specular 8 %, ring 7 % on P1 (ours). "Dominant failure mode" removed.
F11	Accepted	"Every/all trends" replaced by "the FoV and initialisation findings"; objective, normal and pose ablations stated as P1-only.
F12	Accepted	LPIPS trade-off stated in abstract, contributions, results, claims and conclusion (0.189 vs. 0.178).
F13	Accepted	Sample s.d. ($n-1$) stated; per-seed paired PSNR differences given; no significance claims.
F14	Accepted	Rendering equations now give $\hat{w} = \sum a_k$ and $\hat{D} = \sum a_k z_k / \hat{w}$, the zero-opacity rule and the expected-depth interpretation.
F15	Accepted	Moments defined as m_r, $r \in \{1, 2\}$, with clamping and ϵ.
F16	Accepted	"At most 327 680", with the observed count for frame 801 (all pixels valid and unmasked) and initial/final counts distinguished.
F17	Accepted	K_{render} and K_{tri} distinguished with principal points and offsets; the untested effect stated.
F18	Accepted	Wording changed to a unit conversion consistent with the poses; MoGe-2 per-frame scale CV reported (12.8 % on P1).
F19	Accepted	Wording corrected; candidate explanations for 0.093 vs. 0.165 listed without choosing one; metric errors now reported.
F20	Accepted	Scale-invariant and scale-dependent terms listed (pseudo-normals and depth TV are not invariant); the image-TV coupling stated.
F21	Accepted	Masks defined for each metric (stereo disparity > 1, finite positive prediction, prior validity for the uncertainty benchmark).
F22	Accepted	G defined as a ratio of means (Eq. in Section III-K); the mean of per-frame ratios differs by at most 5.2 points and is mentioned.
F23	Accepted	Decile plot renamed a ranking diagnostic; "calibration" removed.
F24	Accepted; recomputed	Fixed precedence device > specular > edge > tissue; all mask parameters and the border caveat stated.
F25	Accepted	Cost split into preprocessing, training, rendering and uncertainty; MoGe-2 4.5 FPS serial cost and ensemble/extra-pass costs stated; unmeasured memory claim removed.
F26	Accepted	Setup states the code commit (39c6ce2), Depth-Anything ViT-B vs. the paper's Small, released loss weights and local modifications; the "unreleased frame range" footnote corrected.
F27	Accepted	SSIM shown with four decimals from unrounded means; difference +0.001.
F28	Accepted	Published SSIM changed to 0.850 (arXiv v4, Table 2).
F29	Accepted	Published numbers labelled as protocol-dependent context; the spread-as-threshold argument removed.
F30	Accepted	Seed s.d. 0.0004 quoted; table prints <.001.
F31	Accepted; new table	Prior-only evaluation (Table V) for both priors and sequences (`stereomis/eval_priors.py`): raw scale and CV, aligned AbsRel, median relative error, valid fraction. It also shows that the trained models inherit the prior's accuracy.
F32	Accepted	Comparators named for every percentage (e.g. 61 % vs. the reproduced baseline, 154 % vs. ours; regional changes vs. FoV-corrected DA); P2_5 ties explained with unrounded values.
F33	Accepted	All listed phrases replaced.

ID	Status	Change
F34	Accepted	"Weak and inconsistent ranking" and "strongest tested individual signal"; relative-ranking caveat added.
F35	Accepted	Novelty bounded to "among the works reviewed" with the search cutoff; established ingredients acknowledged; uncertainty literature split by target; compatibility with other architectures stated as untested.
F36	Accepted	Table I uses wrapped columns at readable size and covers both protocols; Table IX (formerly VII) has one row per source with FPS and GPU.
F37	Partly	Reference numbers removed from figure graphics (qualitative and per-frame figures regenerated). Author names remain placeholders until the authors provide them.
F38	Accepted	PRIMU description corrected (colour and depth; generalisation to unseen scenes); FisherRF placed with depth-AUSE work.
F39	Accepted	Corrections applied to the source fragments; the build script checks return codes, halts on errors, rejects undefined references and unfilled placeholders; empty TikZ `textwidth` keys replaced by explicit styles.
F40	Noted	Builds cleanly; target venue and page limit are not yet specified, so compliance cannot be checked.

Open items that require new experiments

Leave-one-component-out training for the objective (F05); objective and pose ablations on P2_5 (F11); an independent geometric reference such as phantom data (F02, F08); additional sequences (F13).